

Dev Gandhi

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[Github](#) | [Linkedin](#)

Education

Indian Institute of Science, Bangalore

2023 - Present

Ph.D. in Computer Science and Engineering

Relevant Courses: Computer Systems Security, Design and Analysis of Algorithms, Computer Architecture, Distributed Systems, Operating Systems, Game Theory

Rashtriya Raksha University, Gandhinagar

2019 - 2023

B.Tech. in CSE with specialization in Cyber Security

GPA: 8.48/10

Relevant Courses: Operating System, Computer Network, Cyber Forensic, Web Security and Vulnerability Assessment

Publications

[ACSAC'25] Non-Bare-Metal User-Space Control-Flow Attestation, *Nikita Yadav, Hrushikesh Salunke, Dev Tejas Gandhi, Vinod Ganapathy.*

[ATC'26] ESROS2: Enhanced Application Sandboxes for Secure ROS2, *Dev Tejas Gandhi, Vinod Ganapathy.*

Projects

System Call Sandbox

- Implemented a per-process kernel-level system call sandbox using eBPF.
- Extracted system call policies through static binary analysis with Angr and enforced them at runtime.

Optimizing Performance of Dilated Convolution

- Optimized dilated convolution using loop unrolling, strength reduction, SIMD, and redundant computation elimination.
- Developed a multithreaded implementation using pthreads for parallel execution.

eBPF User-Space Stack Walker

- Built an eBPF-based user-space stack walker for kernel-level runtime stack inspection.
- Recovered user-space call chains through stack-frame analysis, similar to GDB's bt.

MLIR Anticipated Expression Hoisting

- Designed an MLIR optimization pass for anticipated expressions using custom dataflow analysis.
- Hoisted redundant computations to improve generated code efficiency.

Experience

Teaching Assistant

Aug 2024, Aug 2025 – Present

- Assisted/assisting in teaching and grading assignments for **Computer System Security**.
- Evaluated student projects and provided feedback.

Presentations

- **ARM Confidential Computing Architecture** — Talk at CSA Summer School 2025.
- **SROS2 Sandboxing Mechanisms** — Presentation at Databased (IISc UG Club).
- **Internals of PyTorch 2** — Presentation for the Advanced Compilers Course.

Skills and Achievements

Achievements

- Achieved AIR 60 in GATE CS'23.

Languages C, C++, Python, SQL, JavaScript, Java, C#

Technologies & Tools eBPF, SELinux, AppArmor, ROS, gRPC, BurpSuite, Linux Kernel, Docker, Spring Boot, Kubernetes, WebAssembly, LLVM, MLIR, PyTorch2.